

Varun Hariharan

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EDUCATION

Northeastern University Master of Science in Artificial Intelligence	Sep 2026 – May 2028
University of Washington Bachelor of Science in Human Centered Design & Engineering; Bachelor of Science in Economics	Sep 2022 – Jun 2026

PROJECTS

Monte Carlo Portfolio Risk Simulator (Python, NumPy, Streamlit, Claude API) - Led end-to-end design of a portfolio risk analysis tool, translating an ambiguous "how risky is this investment" question into a structured, quantifiable framework (VaR, CVaR, probability of loss). - Modeled up to 10K correlated market scenarios using Geometric Brownian Motion, historical covariance matrices, and Cholesky decomposition; calculated VaR, CVaR, percentile ranges, and probability of loss. - Validated model reliability through a rigorous backtesting framework across 1,000+ historical data points, using statistical hypothesis testing (Kupiec test) to confirm forecast accuracy within 0.6 percentage points of target, which confirmed that the model is well-calibrated and trustworthy for decision-making use.	August 2026 – Present
AI Data Analyst (Python, FastAPI, Streamlit, Claude API) - Designed a self-serve analytics tool enabling non-technical users to extract data-driven insights without writing code, reducing the technical barrier between raw data and business decision-making. - Built a multi-step reasoning system capable of autonomously determining what analysis was needed to answer compound business questions, then synthesizing results into clear, actionable, natural-language recommendations. - Integrated Claude-generated findings grounded in computed analysis results, producing natural-language explanations, statistics, and visualizations from executed analysis tools.	Jun 2026 – Present
Research Clipper (Python, FastAPI, PostgreSQL, pgvector, Claude API) - Co-led development of a research tool addressing a specific reliability gap (unverifiable AI-generated claims), prioritizing trustworthiness over convenience in the core design decision. - Rigorously validated system accuracy through a structured 28-question test framework spanning multiple failure modes, achieving 100% accuracy, this included correctly identifying when insufficient evidence existed rather than guessing.	Jul 2026 – Present

EXPERIENCE

CoMotion Labs — Patent Quest (Undergraduate Capstone) (Figma, Prototyping, Playtesting) Product Developer - Translated abstract IP and patent strategy concepts into intuitive gameplay mechanics for early-stage innovators, balancing technical accuracy with user comprehension. - Took on product management responsibilities for several weeks, using playtesting data to identify decision-flow friction points and prioritize the team's next iteration.	Jan 2026 – Jun 2026
Valorant Registered Student Organization (Figma, Accessibility Design, Usability Testing) UI/UX Designer - Identified the highest-leverage problems affecting a 1,500+ person community through direct stakeholder research, then prioritized solutions by expected impact rather than ease of implementation. - Prioritized 4+ rounds of product iteration based on observed user behavior and usability evidence.	Jan 2025 – Apr 2025

PRODUCT & RESEARCH

Seesaw (Sep 2024 – Mar 2025): Goal-tracking app for UW students; research with 25 participants, 8-user usability testing
City (Sep 2024 – Dec 2024): Digital festival experience; branding, competitor analysis, mobile app concept

SKILLS

Data & Quantitative Analysis: NumPy, Pandas, Matplotlib, Plotly, Risk Modeling, Monte Carlo Simulation, Statistical Backtesting, SQL (PostgreSQL)
Programming: Python, Bash/Shell scripting, Java, JavaScript, R, HTML/CSS, React
AI/ML: Claude API, RAG, Agentic AI, Tool Use/Function Calling, Prompt Engineering, Sentence Embeddings, pgvector, Vector DB, PyTorch
Tools: FastAPI, Streamlit, Git, Figma, VS Code, Vercel
Product & UX: Product Thinking, User Research, Usability Testing, Prototyping, User-Centered Design, Competitive Analysis